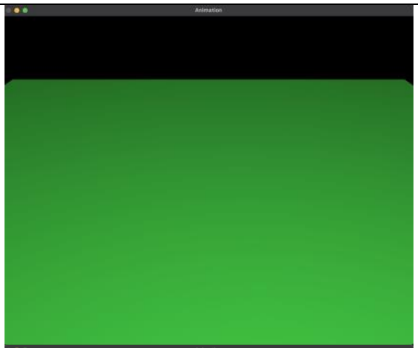
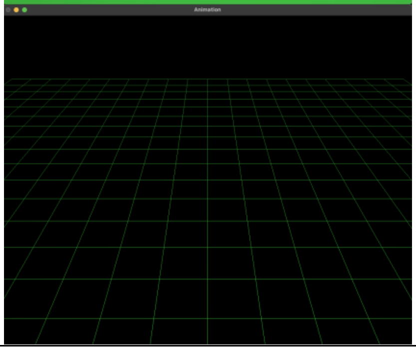
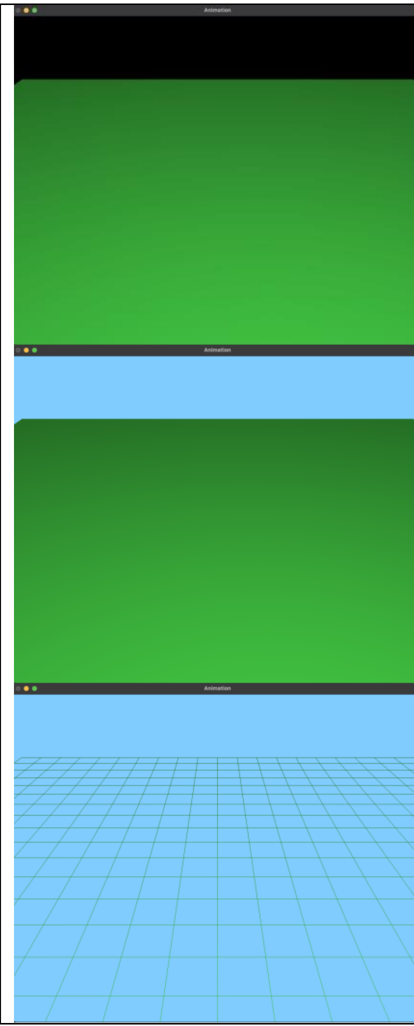
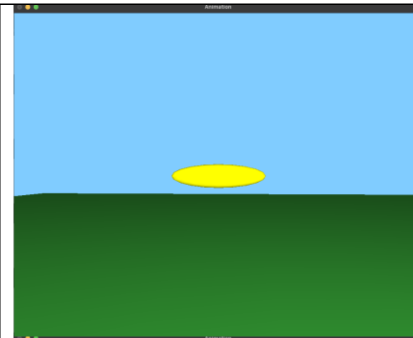
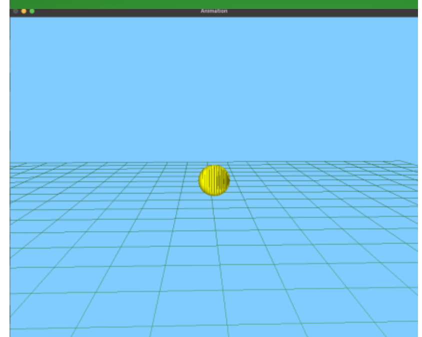
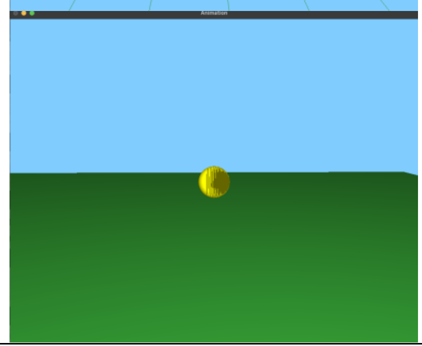
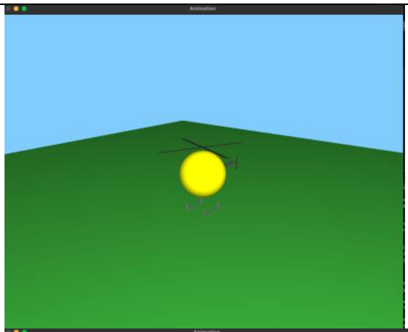
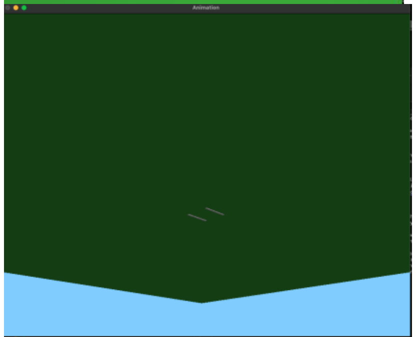
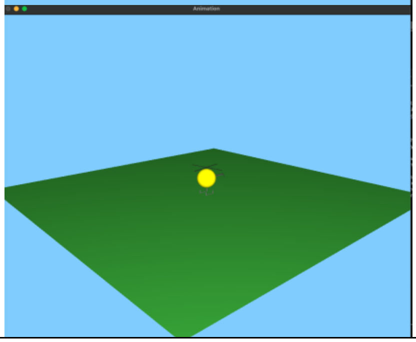
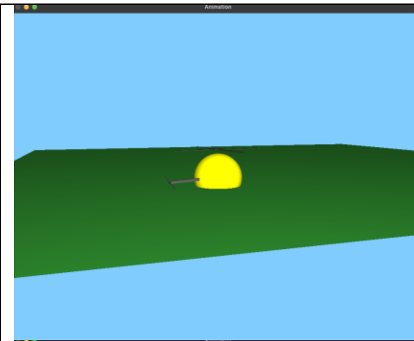
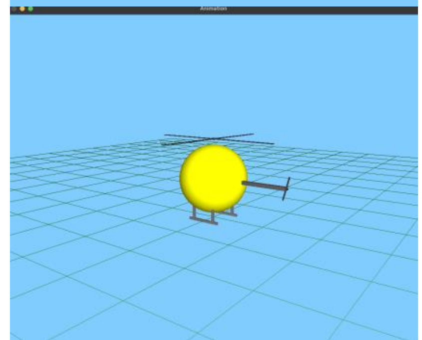
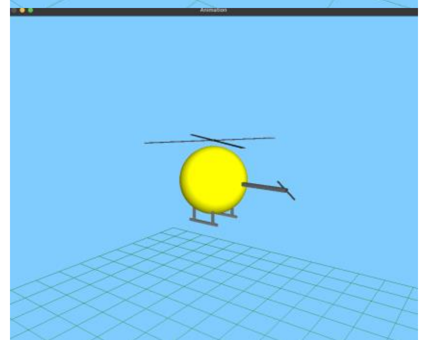
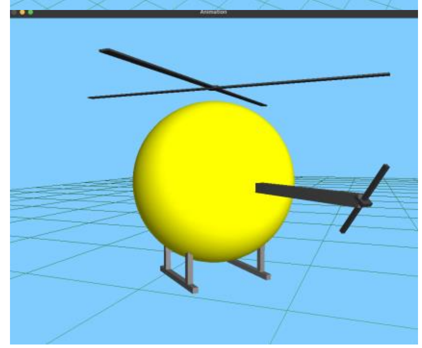
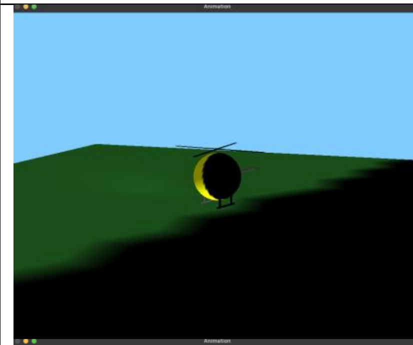
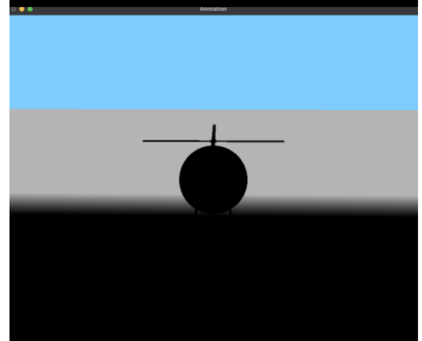
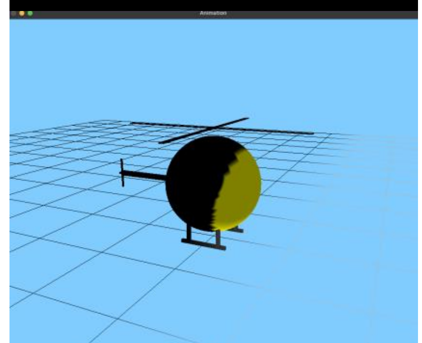
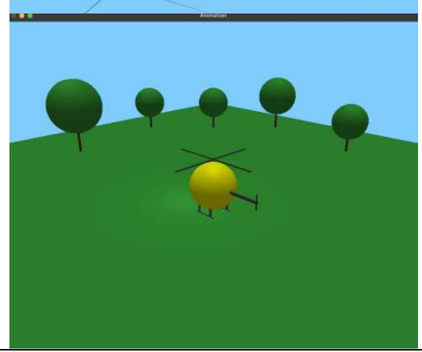
Duration	Date	To Do	Completed	Detail	Issues/Bugs
1 Hour	30-09-2025	- Figure out how to develop on Mac (code will later be converted for VS on windows)	Able to run program on VSC, will have to transfer everything over to VS2022 once I have access to a windows device.	n/a	n/a
1 Hour	01-10-2025	- Start with drawing a basic ground. - Toggle wireframe - Set up camera perspective - Ensure backspace bug from snowman doesn't re-occur.	Ground drawing completed, still using the provided lighting but will change that later on. Wireframe successfully toggled Camera perspective set up Backspace bug avoided by using ESC key to close the program rather than terminating it via terminal. (As far as I know this issue is only on Mac OS, so it may not be noticed at all on windows but good practice to fix bugs).		Minor issue with camera setup due to type within gluPerspective

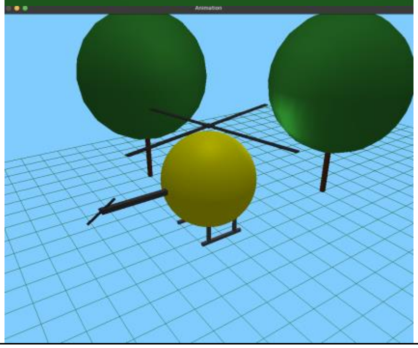
2 Hours	02-10-2025	- Basic scene settings, and rational for chosen scale. - Add sky color	1.0 GL unit = 10.0 meters. After quick research, it is safe to conclude that the average helicopter is around 10 to 15 meters long as well as the diameter of its blades being of a similar number. Therefore, following my chosen 10 meter/unit, it makes the model easy to work with. Likewise, trees, ground, lakes, and even camera positioning will all be measured in Meters, once again making “Meter” a good choice to rationalize my chosen scale. While Centimeter was considered due to fine tuning, its way too unnecessary for the current project, and simply 0.1m can be used in rare instances instead of 10.0cm. This is why 1.0 GL unit = 10.0 Meters for my project and a helicopter scene.		Sky not changing color despite correctly writing glClearColor in init(). Maybe something is clashing? But I doubt it this early on. Turns out, I’ve forgotten to Recompile before running the program (always the simplest mistakes).

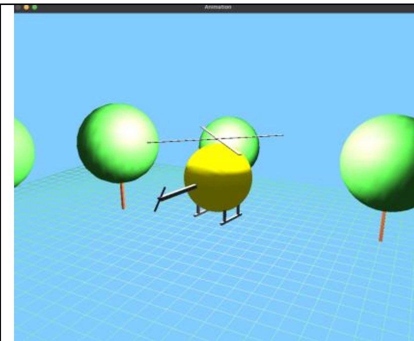
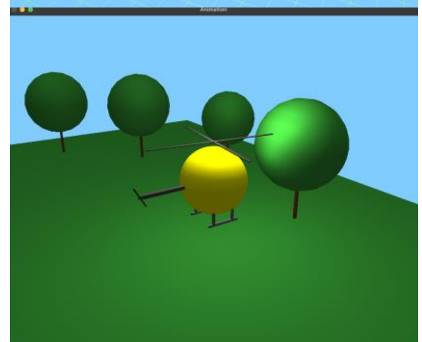
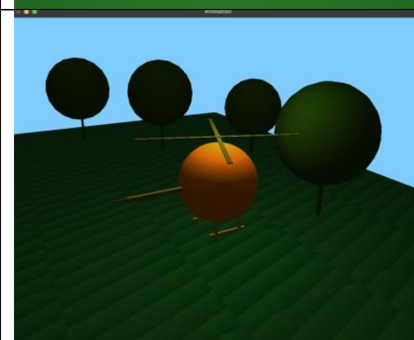
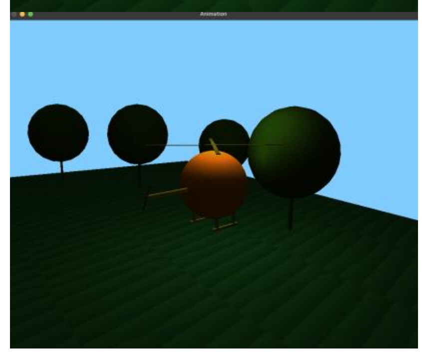
		- Camera control - Lay basic shapes for Helicopter		   	I'm developing on MacOS and a slight hiccup ive ran into, is the lack of "glutMouseWheelFunc" which simply means I can't zoom in/out by scrolling the mouse wheel. While this isn't a serious issue, and I can probably bind it to a button on the keyboard, its worth noting. Currently, my helicopter fuselage looks like a beautiful lemon. Which is why OpenGL lighting may be struggling with it. I'll change it more to a "sphere" and see if there is an improvement. Turned my lemon to a sphere, yet the lighting is still harsh at a 90-degree angle. Simply increasing the number of vertical and horizontal slices to double my previous value improved it, but may slow down rendering. As the helicopter will be the focus of this project, I've decided that it is worth dedicating more triangular faces towards it compared to other items that will be present later in my project.

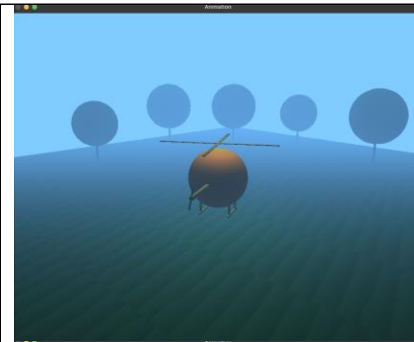
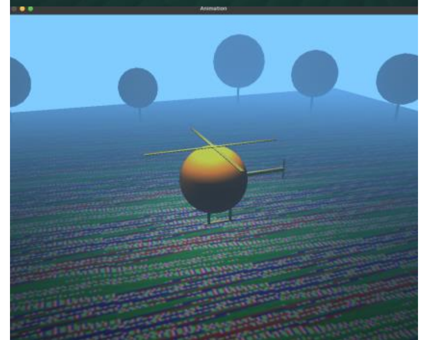
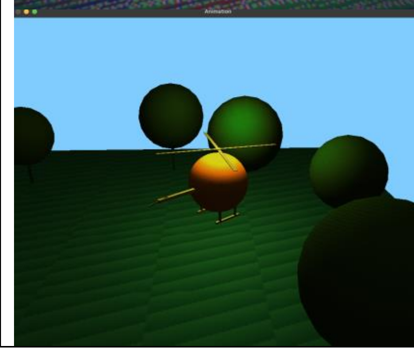
3 Hours	15-10-2025	- Depth test needs to be done, might fix lighting issue? - Propellers + landing gear to helicopter. - Add key-binds for zoom button -	GREAT SUCCESS!!, Simple GL_DEPTH_TEST fixed my lighting issue and the helicopter isn't stuck to the camera anymore!!. In the 2nd picture you can see that the helicopter is staying above the ground when viewed from beneth the ground (if that makes sense). Basically, the layers are working correctly. Due to an issue with MacOS, I had technical errors getting the zoom to work with the scroll wheel. Work around was to use "I" for zoom in, and "O" for zoom out. Only used OpenGL matrix transformation as per requirements.		Rear propeller took forever to get right, turns out I had to rotate it from the Z axis rather than the X and Y axis. Not sure why it took me so long to figure that out.

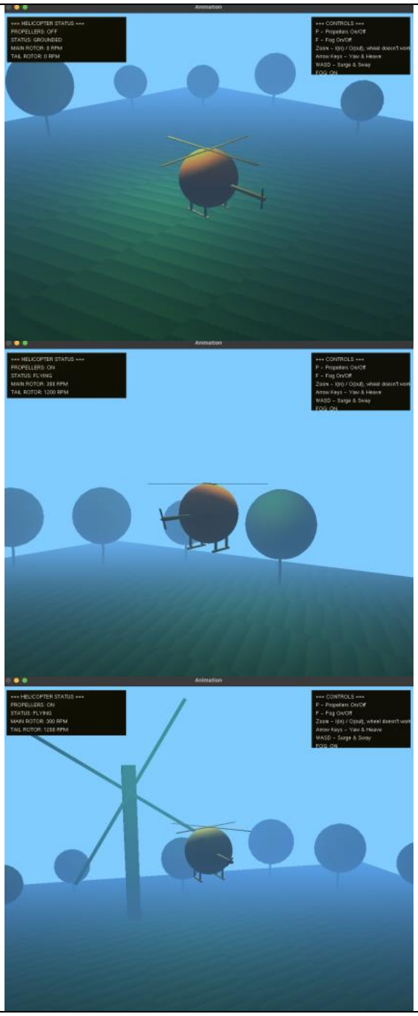
2 Hours	17-10-2025		- Movement - Ground collision with landing gear - Have the Helicopter fully controllable.	works great! several inputs can be made at the same time with the helicopter following them with no issues caused to the program. Ground collision with the landing gear successful.	   	The helicopter moves but as soon as I turn in any direction, the motion goes the opposite way. E.g The helicopter faces right, but flies left. My movement calculation wasn't matching the visual rotation, classic error in my trigonometric function caused this. Correcting my sin and cos for calculating movements and, doing some research to better understand openGL's coordinate system helped resolve this. Sadly this took longer than what I initially wanted (1 to 2 hours maximum, spend 3 hours). Ground collision is causing some issues again. At high speeds, helicopter would pass beyond the Y limit set due to no collisoin detected. I aimed to set a hard collision detection with position forcing, but I faced inconsistend landing heights. While my collision was set to 3.5f, the helicopter landed higher naturally. When it finally did land on the "ground" it would freeze and be unable to take off. My collision detection condition caused a circular dependencie. Naturally, the solution would be a buffer zone right?. This was much more complicated than I expected, due to my own poor planning. First I had implemented simple ground collision at Y = 3.5f, but the helicopters landing gears still slightly penetrated the ground inconsistently depending on the helicopters speed, almost as if the helicopter is "sinking". I then tried a hard collision, forcing the position of Y = 3.51f, to prevent the ground penetration issue, but then I couldn't take off anymore. A buffer system sorted this out and also a debug output in the terminal to make sure the hard collision was working regardless of speed.

					(yes, I know the propellers need to be adjusted, was thinking of adding a rectangle that connects it to the fuselage but might just drop the Y axis of the propellers and attach it to the fuselage itself).
7 hours	18-10-2025	- Lighting - Make an early concept of lighting - Make the lighting work (sun + spotlight on helicopter).	Added a lighting to fuselage of the helicopter (pretending its just the illumination inside the cockpit). Sun working successfully. (directional light) Spotlight is working. Added some basic trees to the environment.		Learning the ropes with lighting, I want to put a spotlight on my helicopter, maybe even an illumination as if a light is coming from the cockpit. The directional lighting of my sun is coming from an angle, which makes it not a very effective sun. My solution was to have the light coming straight down which seem to fix the problem. At first the spotlight (GL2) was too dim, so I played around to fine tune it. Played around with the diffuse and specular levels until I was happy with the outcome. The spotlight had some issue over distance, which made it useless as a spotlight. Adjusting attenuation settings made it stronger over distance. From the very bright image you can see I set it to a very high number just to see what it's capable of. This made me learn how to make a neon-like effect. Further fine tuning and I found a level that I was satisfied with the distance and strength of my spotlight.

				 	
4 Hours	20-10-2025	- Apply textures - Apply fog		 	Unfortunately, after applying textures it was dim and needed re-tuning. And then after adding fog, it once again needed to be re-tuned. The diffuse and specular were changed several times to reach a satisfying level. Attenuation was also adjusted to keep the beam stronger over longer distance, so the helicopter didn't need to go up against a tree for the spotlight to be seen. My ground texture had gotten corrupted at one point, I presume because I had accidentally opened the texture file it caused the corruption? Thankfully I had a backup saved and just restored the texture. Added a toggle button (F) to turn fog on and off just to see if the light has improved or if its over-done.

2 Hours	22-10-2025	- Add a HUD showing controls + helicopter status. - Add an animate object in the map.	Added a HUD, showing the Helicopters status (flying/grounded). Showing the propellers RPM increasing, until it reaches a sufficient amount to permit taking off. Added a simple windmill as an animated object.		Noticed I still have GL color enabled, changed it to GL material. Sadly this got rid of the two-tone color my helicopter that I liked, sure its another issue with the order of the coloring but that isn't a main priority at the moment.